

# Fundamentals of Photovoltaics (EE336)

Join QQ group

### Class Venue

Lecture: Room 304, Teaching building 1, 10:20-12:10

Lab : Engineering Building – South Tower . Room 322

### Instructor:

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### Lab & TA:

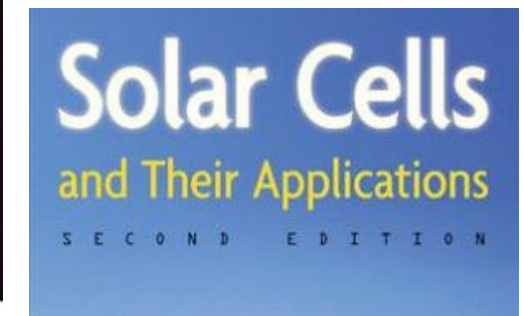
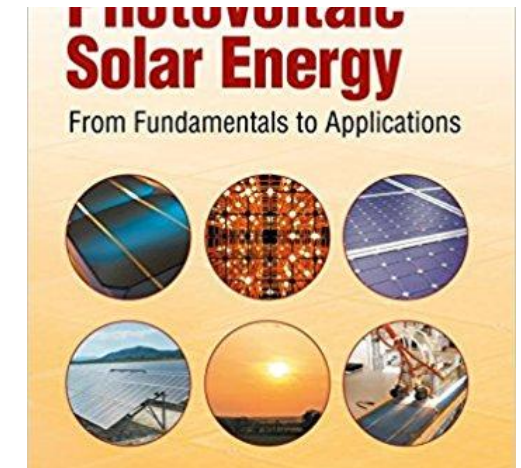
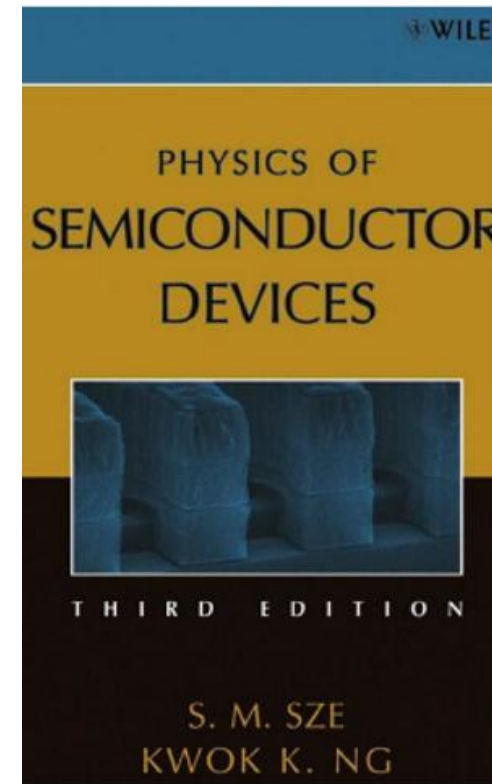
刘荣辉 Liu Ronghui: [12431232@mail.sustech.edu.cn](mailto:12431232@mail.sustech.edu.cn)



# Reference Books and Materials

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- We will mainly focus on lecture notes. However, if you want to read more, refer to the following books and materials.
- S. M. Sze and Kwok K. Ng, Physics of Semiconductor Devices, Chapter 13 (Photodetectors and solar cells)
- Lewis Fraas and Larry Partain, Solar Cells and Their Application
- Angele Reinders, Pierre Verlinden, Wilfried van Sark, Alexandre Freundlich, Photovoltaic Solar Energy: from fundamentals to applications, ISBN 9781118927465
- [www.pveducation.org](http://www.pveducation.org)



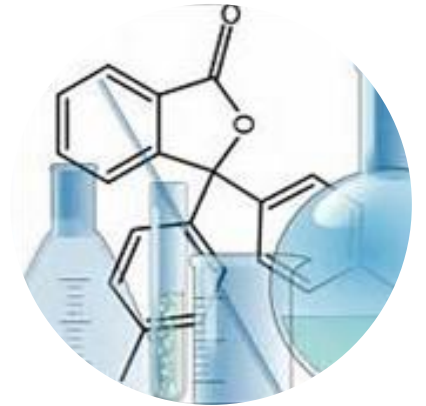
# Tentative Schedule (Lecture)



|                         |                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Section 1               | Introduction (environmental issue, energy demand, overview of photovoltaic technologies, history, status, trend and future) (Week 1) |
| Section 2               | Basic of light and solar spectrum (Week 2)                                                                                           |
| Section 3               | Semiconductor Materials and Their Properties (Week 3-4)                                                                              |
| Section 4               | Operation Principle of Photovoltaics (Week 5-6)                                                                                      |
| Section 5               | Silicon Solar Cells (Week 7-8)                                                                                                       |
| Mid-term Exam (Week 9 ) |                                                                                                                                      |
| Section 6               | Organic Solar Cells (Week 10-12)                                                                                                     |
| Section 7               | Dye-sensitized Solar Cells (Week 12-13)                                                                                              |
| Section 8               | Perovskite Solar Cells (Week 14-15)                                                                                                  |
| Section 9               | Revision (Week 16)                                                                                                                   |

# Tentative Schedule (Lab)

- Lab 1 : Sun Position Calculator
- Lab 2 & 3 : Fabrication of Perovskite Solar Cell
- Lab 4 : Characterization of Perovskite Solar Cell
- Lab 5 : Fabrication of Organic Solar Cell
- Lab 6 : Characterization of Organic Solar Cell



# Assessment

- Lab (30%)
- Mid-term test (20%)
- Final Exam (40%)
- Class Participation (10%)



## Why should you choose this course?

- ☐ Renewable energy is a major need for the 21<sup>st</sup> century
- ☐ Learn all types of photovoltaic technologies under one course
- ☐ Acquire both theoretical knowledge and hands-on experience
- ☐ > 80% of the class got a grade B and above
- ☐ No one has failed so far